

Liquid – Liquid Extraction

Pre-Lab Plan
Unit Operations Lab #3
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Group 3, Tuesday, PM section

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Commented [T1]: 31.5/35

1. Experimental Objective (5 pts) Alexis

The objective of this experiment is to study the effects that agitation speed has on the efficiency of solvent extraction. A three-stage mixer-settler system will be used to separate acetic acid from white oil using water as the solvent. The system variables that will serve as independent variables for this experiment include the speed of the agitator, and the flowrates of the white oil-acetic acid mixture and water.

2. Experimental

In this experiment, a three-stage solvent extraction unit will be used to separate acetic acid from an oil-acetic acid mixture using water as the solvent. The oil-acetic acid mixture used will be 0.5 wt% of acetic acid in oil. Water will flow from a holding tank on the far left of the apparatus in a countercurrent direction to the oil-acetic acid mixture which will flow from 25 gallon holding tank on the far-right side of the apparatus. Agitators will be utilized in each of the three stages to maintain well mixed conditions.

2.1 Apparatus

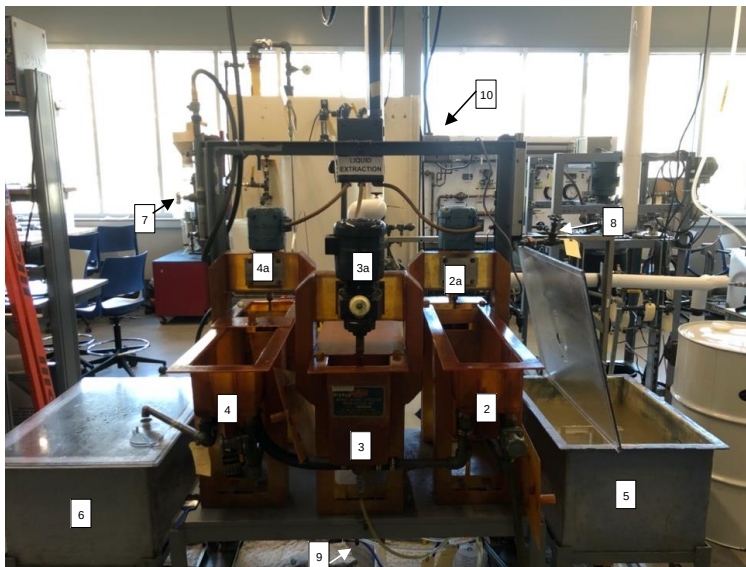


Figure 1: Apparatus of liquid-liquid extraction

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Mention that you'll use McCabe-Thiele analysis

Commented [AZ3]: 'of a liquid-liquid extraction process'

Commented [T4]: 4/5

Missing pump, agitator power switch, pump switch

Table 1: Apparatus components and description

Component	Description
1	Impeller mixer to ensure uniform mixing of components
2	Extraction Unit Stage 1
2a	Agitator for stage 1
3	Extraction Unit Stage 2
3a	Agitator for stage 2
4	Extraction Unit Stage 3
4a	Agitator for stage 3
5	Feed Tank (Mineral Oil-Acetic Acid Mixture)
6	Raffinate Tank
7	Water Flow Control Valve
8	Feed Tank Control Valve
9	System Drain Valve
10	Centrifugal Pump to deliver contents from feed tank to extractor system

2.2 Materials and Supplies

Table 2: Materials and supplies used in experiment

Materials	Description
White oil	Mineral oil mixed with acetic acid and fed into a mixer. Volume of 25 gallons and 0.5 wt% or acetic acid.
Acetic acid	Mixed with white oil and fed into a mixer, tank located on the right side.
Water	Used as solvent and located in left large tank.
Standardized NaOH	Used in titration to measure composition of acetic acid.
Phenolphthalein	Indicator used in titration.
Safety gloves	Protects hands from any chemicals present in the lab.
Safety goggles	Protect the eyes from chemical splashes.
50 mL burette	Holds standard NaOH solution for titration.
Burette clamp	Holds burette in place.
125 mL Erlenmeyer flasks	Contains liquid samples for titration procedure.
Parafilm	Used to cover the samples that are titrated.
Tachometer	To measure the RPM of the agitators

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Missing stir bar and stir plate

Plastic pipettes and bulbs to collect samples

2.3 Experimental Plan

Before proceeding with the experiment, the Denver system should be inspected carefully to ensure that all lab members understand the unit and how it operates. The following are a few key points to know before the procedure:

- Each stage of the unit contains a mixer and a settler and arrangement of reflux
- Agitator speeds are adjusted using a knob on the motor
- Agitator speeds are measured with a tachometer
- The flow rates of the feed and the solvent are measured through the use of rotameters
- The flow rates of extract and raffinate phases are measured with measuring cylinders

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Procedure

1. Prepare a 25-gallon mixture of White oil and acetic acid
 - the mixture should be close to 0.5 wt% acetic acid
 - make sure to do a preliminary titration of the white oil in the tank to determine its initial composition
2. Thoroughly agitate the 25-gallon mixture inside the feed tank
 - this is done through operating the centrifugal pump in a closed circuit
 - make sure not to run the centrifugal pump in dry conditions, this can destroy the pump, the feed tank should always have at least the minimum level of liquid
3. Once the mixture of acetic acid and white oil is completely mixed take a sample and determine the composition of the mixture
 - the composition of acetic acid in water or white oil is determined through titration
 - titration is performed against a standardized NaOH solution, with phenolphthalein being used as the indicator
4. Establish the solvent circuit, water in this experiment, in the mixer-settler unit until steady levels are obtained for each stage.
 - keep in mind that the white oil flow cannot be started until the water circuit is established
5. Adjust the speed of the agitator to the correct value
 - the agitator speed should not exceed 1000 RPM or go below 300 RPM
 - agitator speed should be higher than 300 RPM to establish flow circuits of the water and oil in all three stages
6. Once the water circuit is well established the white oil is fed into the system
 - the flow of the white oil should be adjusted to ensure that the ratio of solvent to feed flow rate is at least 2.0

Commented [T7]: How much acetic acid? In ml?

- typical flowrates for appropriate operation conditions include water flowrate of 0.67 GPM and white oil flowrate of 0.34 GPM
7. Continue adjusting the flowrates of water and white oil until steady levels of the two phases are reached in each stage of the apparatus
 - once steady state is reached be sure to keep these flowrates constant throughout the rest of the experiment
 - make sure that the chosen flowrate of white oil allows for three runs to be conducted within the 25-gallon constraint
 8. After steady state of the system is established samples of the white oil and water phases should be taken from each of the three stages
 - once again, the composition is determined through titration as mention earlier in step 3
 9. Repeat the sampling for three different agitator speeds (500, 600, and 700 RPM)
 - make sure to keep the flowrates constant throughout the whole experiment
 - make sure to also keep all reflux valves fully closed
 10. Make sure all compositions of samples, flowrates, and agitator speeds are properly recorded in an excel file for the calculation and construction of the following:
 - plots of acetic acid concentration as a function of stage number and agitator speed (for two phases, oil and water)
 - plot of acetic acid in extract vs. acetic acid in raffinate for different agitator speeds
 - Murphree efficiencies for each stage and the overall efficiency

The independent variables in this experiment are the speed of the agitator and the flowrates of the white oil and water. The dependent variables in this experiment are measured compositions of the samples.

Commented [T8]: And the initial acetic acid concentration in the oil

Project Safety Evaluation (10 pts)

Project Title: Liquid-liquid Extraction

Date: 2/18/2020

	Yes/No	
Potential electrical hazards?	yes	If yes, identify conditions: water splashing onto the motor and the pump could cause electrical smoke or fire
Potential mechanical hazards?	yes	If yes, identify conditions: excessively high motor speeds (>1000 RPM) could splash chemicals onto equipment and bodies
Potential pressure hazards?	no	If yes, identify conditions:
Potential temperature hazards?	yes	If yes, identify conditions: Motor or pump could become too hot and burn skin if touched
Hazardous/toxic chemicals involved?	yes	If yes, hazards involved: acetic acid – toxic, corrosive to skin and eyes, and flammable Sodium Hydroxide – corrosive to metal and causes skin burns and eye damage
Gloves required?	yes	If yes, specify type: nitrile gloves
MSDS reviewed by all team members?	yes	List MSDS sheets reviewed: white oil, acetic acid, sodium hydroxide, phenolphthalein indicator and water
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	25	Room temperature water
Pressure (psia)	14.7	System is at atmospheric pressure
Safety Controls	yes/no	If yes, Provide description
1. Pump power switch	Yes	On/Off switch
2. Agitator knob		Knob on the motor that adjusts the mixing speed
3. Reflux valves		Open and close reflux opening
4. Water Valves		Open/close valve to fill tanks with water

I have read relevant background material for the Unit Operations Laboratory entitled: “Liquid-liquid Extraction” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Kellen Krupa 2/18/2020
signature of team member date

Adisa Hadzic 2/18/2020
signature of team member date

Commented [T9]: 10/10

Commented [T10]: Usually that doesn't happen, but OK to mention as a possible hazard

Commented [AZ11]: And skin!

Commented [AZ12]: Also the agitator power switch.

<u>Alexis Miranda</u>	<u>2/18/2020</u>
signature of team member	date
<u>Alice Guzik</u>	<u>2/18/2020</u>
signature of team member	date
<u>Ian Lim</u>	<u>2/18/2020</u>
signature of team member	date